

生成式AI实践

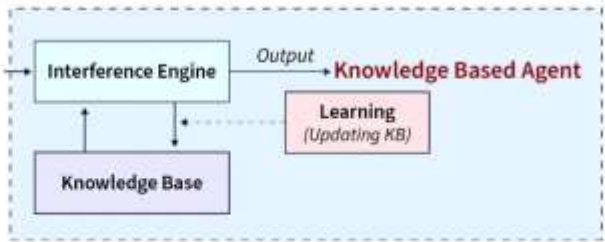
2026春 计算机图形学
Project 3

什么是Agent？

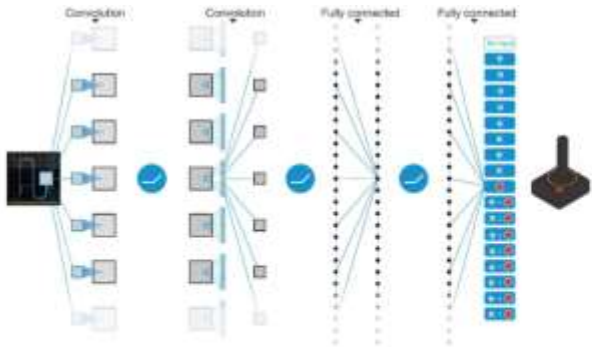
自主性级别 (Autonomy Level)	示例系统 (Example Systems)	解锁的 AGI 级别 (Unlocking AGI Level(s))	引入的示例风险 (Example Risks Introduced)
自主性级别 0：无 AI <i>人类完成所有工作</i>	模拟方法（例如，用铅笔在纸上画草图） 非 AI 数字工作流（例如，在文本编辑器中打字； 在绘图程序中画图）	无 AI	不适用（现状风险）
自主性级别 1：AI 作为工具 <i>人类完全控制任务，并使用 AI 自动化繁琐的子任务</i>	借助搜索引擎寻找信息 借助语法检查程序修改写作 使用机器翻译应用阅读标牌	可能：初级狭义 AI 很可能：胜任狭义 AI	技能退化（例如，过度依赖），对成熟行业的颠覆
自主性级别 2：AI 作为顾问 <i>AI 承担实质性角色，但仅在人类调用时</i>	依赖语言模型总结一组文档 使用代码生成模型加速计算机编程 通过复杂的推荐系统消费大部分娱乐内容	可能：胜任狭义 AI 很可能：专家狭义 AI；初级 AGI	过度信任，激进化，定向操纵
自主性级别 3：AI 作为合作者 <i>人类与 AI 平等协作；目标与任务的交互式协调</i>	通过与下棋 AI 的互动和分析来进行国际象棋棋手训练 通过与 AI 生成的人格进行社交互动来娱乐	可能：初级 AGI 很可能：专家狭义 AI；胜任 AGI	拟人化（例如，准社会关系 / 虚拟恋爱），快速的社会变革
自主性级别 4：AI 作为专家 <i>AI 主导互动；人类提供指导和反馈或执行子任务</i>	使用 AI 系统推进科学发现（例如，蛋白质折叠）	可能：卓越狭义 AI 很可能：专家 AGI	社会规模的精神倦怠 (ennui)，大规模劳动力流失，人类例外论（特殊性）的衰落
自主性级别 5：AI 作为智能体 <i>完全自主的 AI</i> 我们在这里	自主的 AI 驱动个人助手	很可能：卓越 AGI；人工超级智能 (ASI)	目标错位 (misalignment)，权力集中

Morris M R, Sohl-dickstein J, Fiedel N, et al. Levels of AGI for Operationalizing Progress on the Path to AGI[C]. Proceedings of the 41 st International Conference on Machine Learning, Vienna, Austria. PMLR 235, 2024.

什么是Agent？



Logical Agent



Neural Agent

我们在这里

Expressiveness	Low bounded by the logical language	Medium anything a (small) NN can encode	High almost anything, esp. verbalizable parts of the world
Reasoning	Logical inferences sound, explicit, rigid	Parametric inferences stochastic, implicit, rigid	Language-based inferences fuzzy, semi-explicit, flexible
Adaptivity	Low bounded by knowledge curation	Medium data-driven but sample inefficient	High strong prior from LLMs + language use

Mnih et al., “Human-level control through deep reinforcement learning.” Nature (2015)

Agent与图形学？

2D生成

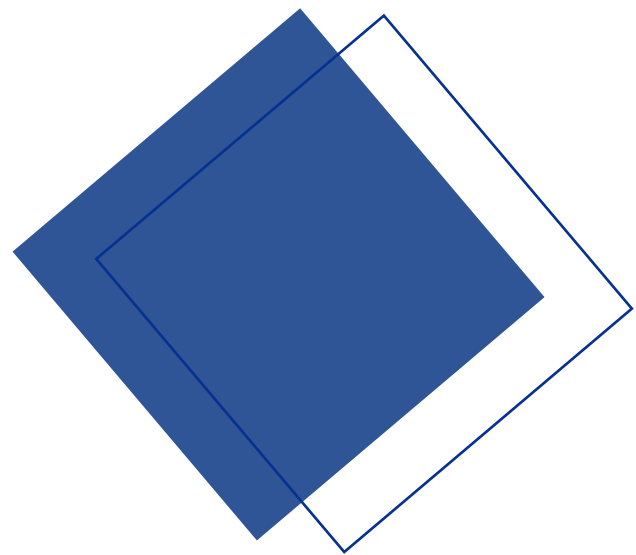
1. 意图解析与自动化，如提示词优化。
2. 精准的2D排版和生成，如海报生成。
3. 多Agent协同协作，如迭代优化。

3D生成

1. 基于NeRF和3DGS的自动生成。
2. 基于Blender的自动化场景搭建。
3. 代码驱动的程序化生成。

视频生成

1. 剧本拆解与分镜规划。
2. 角色与场景的时空一致性控制。
3. 视频素材的理解和剪辑。



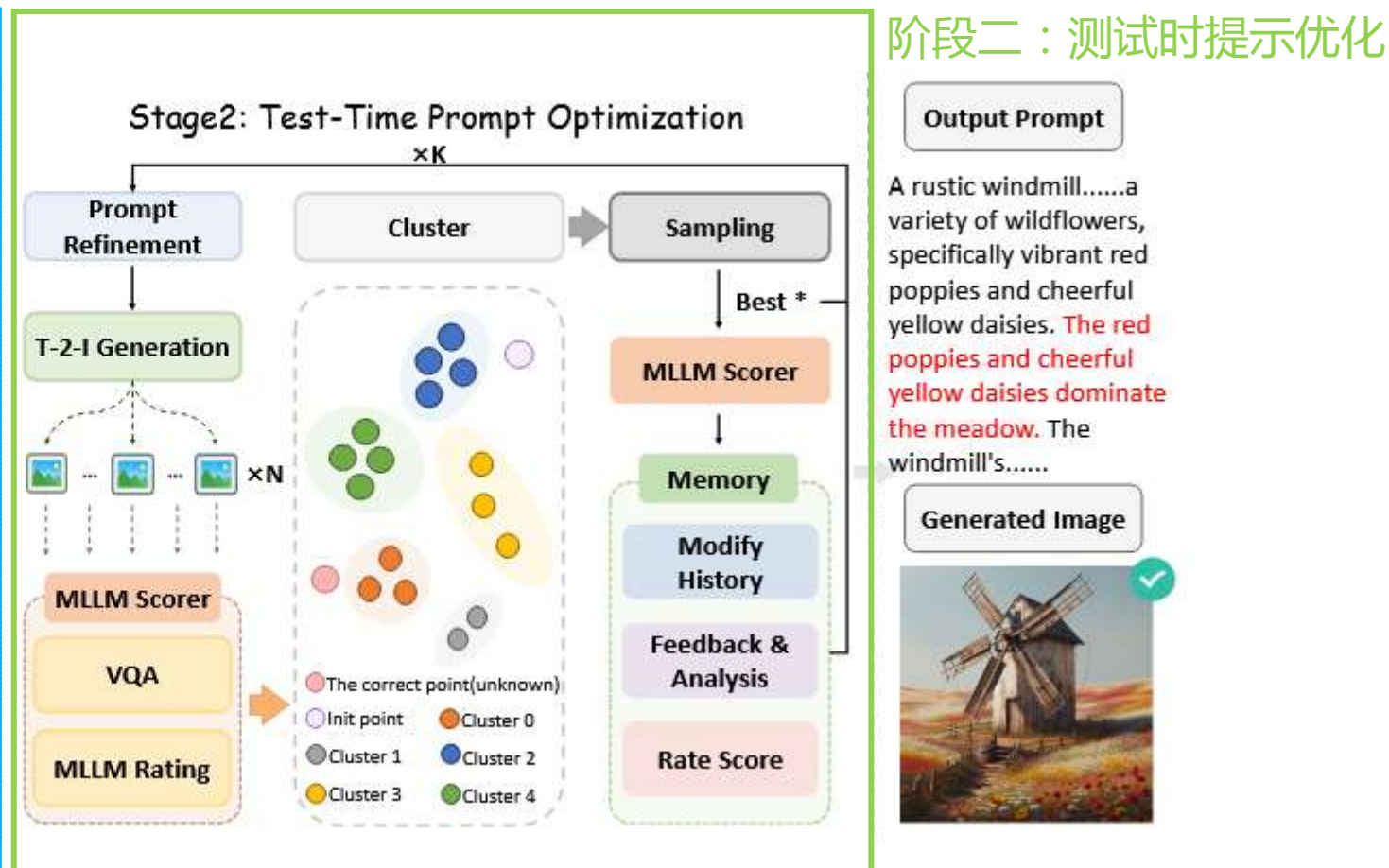
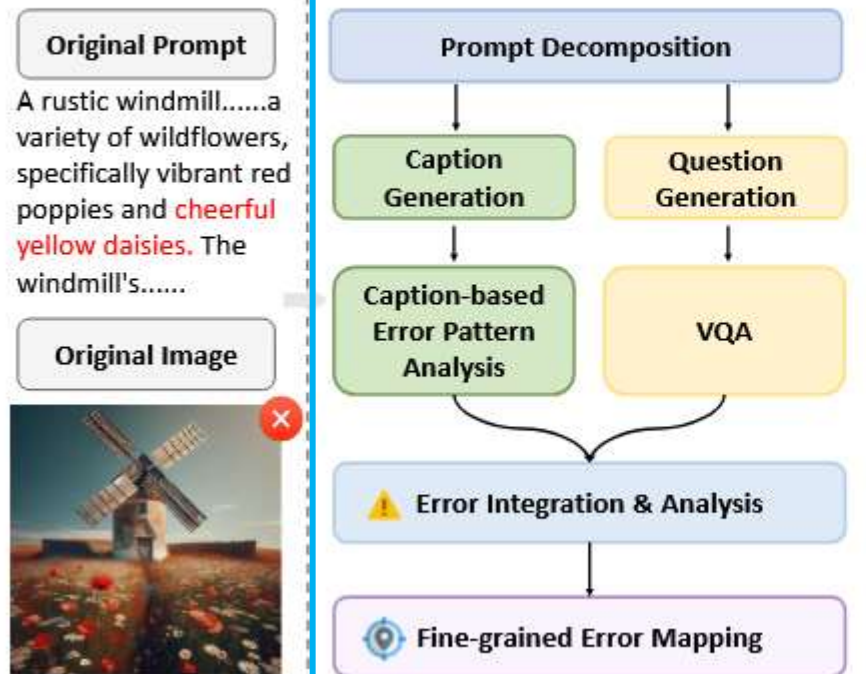
01 2D生成

意图解析智能体 —— GenPilot

文生图模型在面对**复杂或长提示词**容易出现**语义不一致、细节丢失**等问题；**现有优化方法**如微调**需要训练模型并且只适配特定模型**，自动提示优化方法（APO）缺乏系统性的误差分析和改进策略，导致可靠性和效果有限；大多测试时增强只改噪声空间或采样数，**不对提示词本身进行修改**，可解释性差。

GenPilot：多智能体（Multi-Agent）测试时提示词优化系统，在不训练、不微调模型的前提下，自动优化提示词，让文生图更准、更贴合描述。

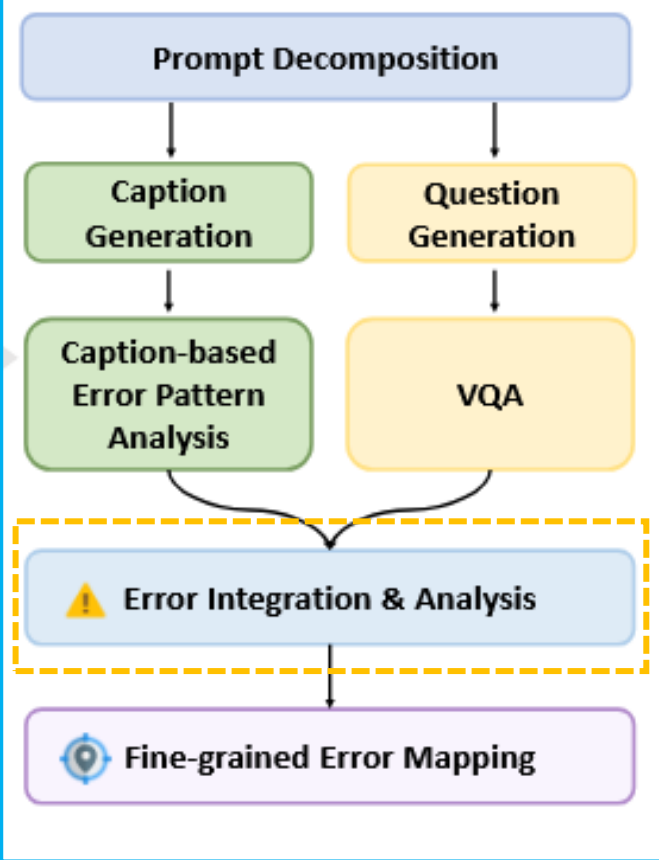
阶段一：错误分析



Genpilot :

阶段一：错误分析

Stage1: Error Analysis & Mapping



提示分解：将原始提示分解为“句子片段” $P = \{s_1, s_2, \dots, s_n\}$ ，每个片段包含对象、关系、背景信息

双路错误检测：

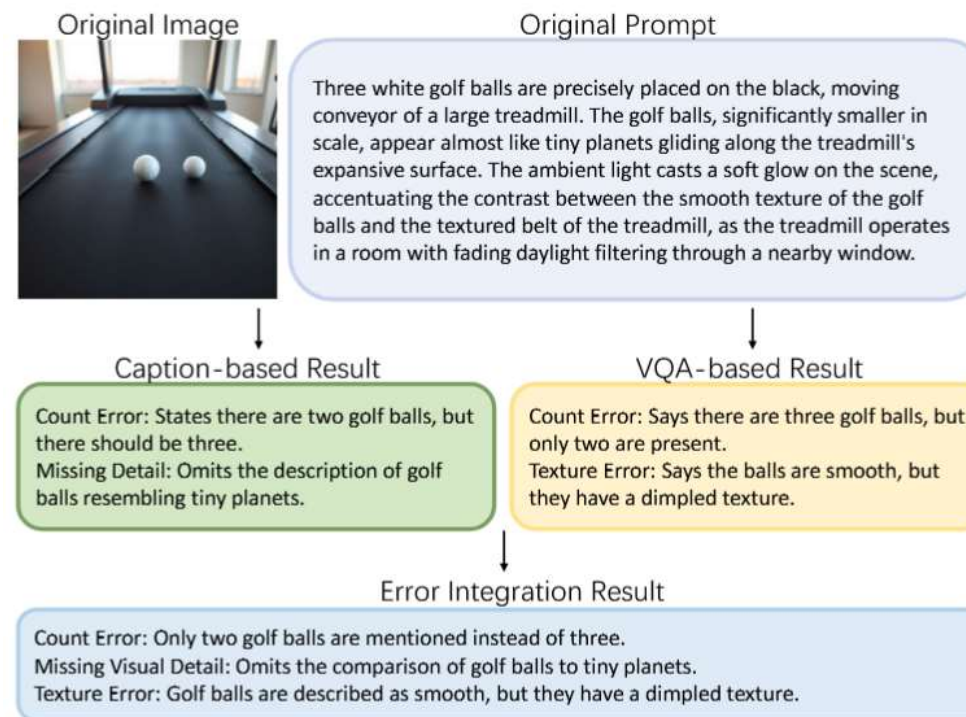
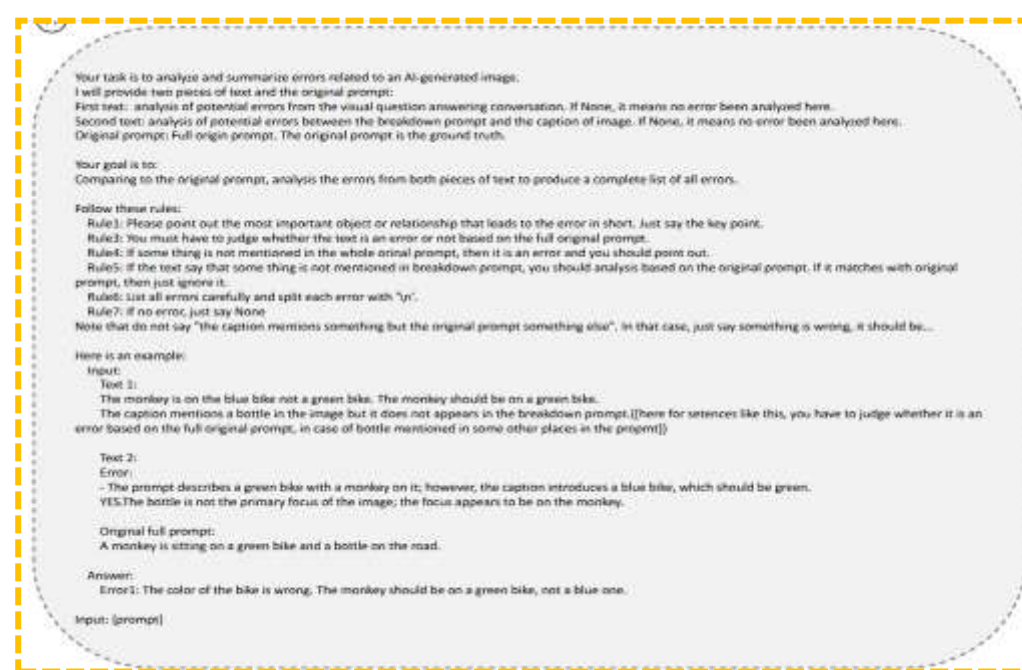
VQA分支：引入了MLLMAgent以生成全覆盖问题。生成Yes/No问题（存在性、属性、空间关系），用MLLM回答

Caption分支：MLLM生成详细描述用于**比较Agent**与原始提示对比

错误融合与映射：

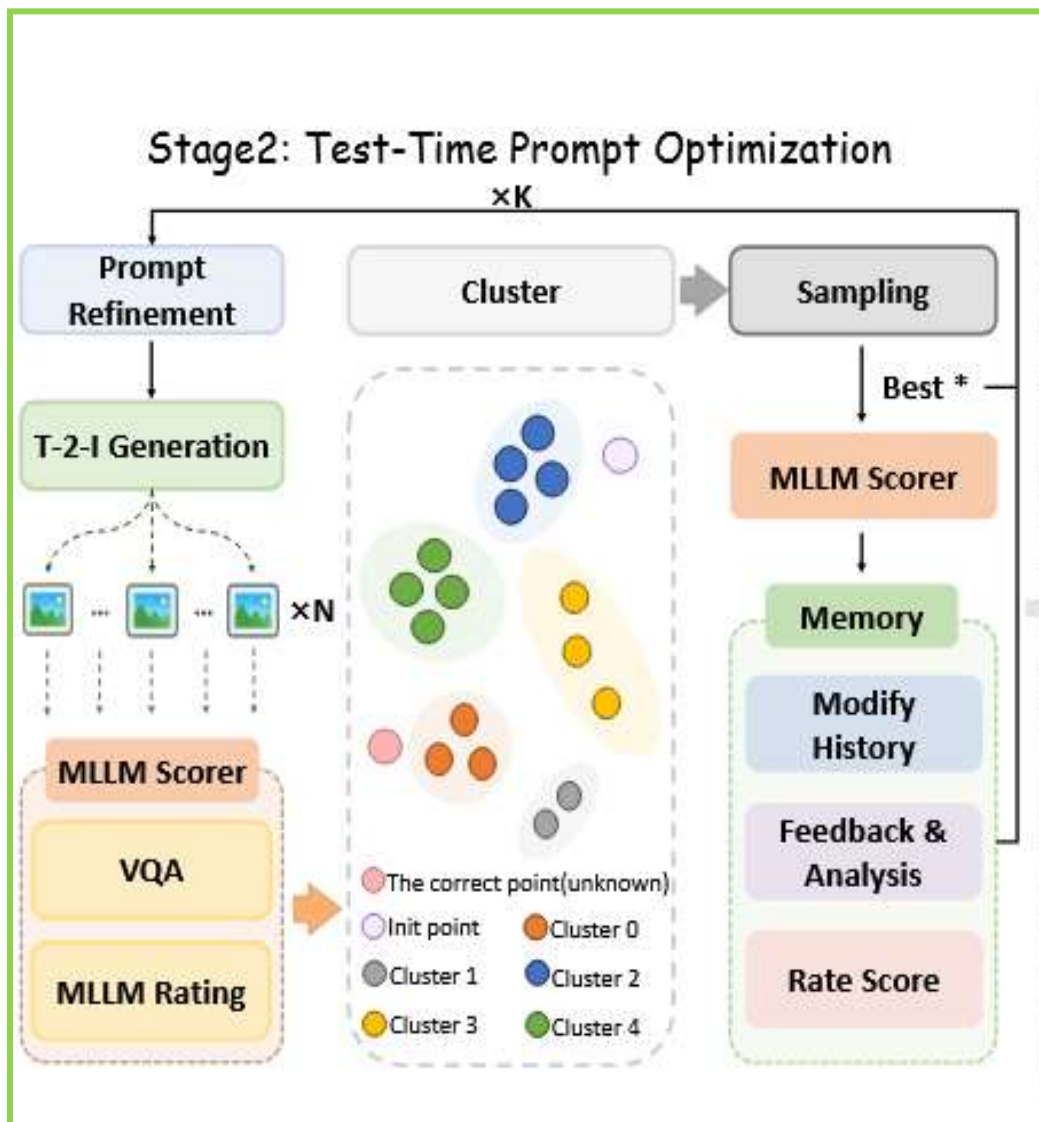
错误整合：Agent错误整合整合两路结果，生成完整错误集

错误映射：将每个错误映射回原始提示的具体句子片段



Genpilot :

阶段二：测试时提示优化



提示词改进：

提示词改进Agent：基于元数据修改错误映射的句子 m_i ，生成多样化的备选改进。

分支-合并Agent：合并 m_i 到原始提示词P中，生成备选提示词 P_i

将备选提示词输入T2IM中生成图像。

MLLM评分器：

VQA Agent：评估不一致性
评分Agent：打分

$$S(P_i^j) = \text{avg}(A_{\text{rate}}(I_i^j, P, A_{\text{vqa}}(I_i^j, P)))$$

聚类：选出最优的提示词集合

记忆模块：存储最优提示、图像、评分、错误摘要，供下一轮参考



Under the soft glow of a rising sun, a round jade-colored table supports six freshly steamed baozi, their white wrappers slightly translucent, emitting tender wisps of steam.



Under the warm and radiant soft glow of a rising sun, a round, distinctly jade-colored table supports **exactly six freshly** steamed baozi, **each with its white wrapper slightly translucent, gently** emitting tender wisps of steam.



Under the soft glow of a rising sun, a round jade-colored table supports **six six** freshly steamed baozi, each with its white wrapper slightly translucent, emitting tender wisps of steam.



Under the soft glow of a rising sun, a round jade-colored table supports **exactly six** freshly steamed baozi, each with its white wrapper slightly translucent, emitting tender wisps of steam.



Under the soft glow of a rising sun, a round jade-colored table supports **precisely six freshly** steamed baozi, each with its white wrapper slightly translucent, emitting tender wisps of steam.

Genpilot :

生成无误差图像



SDv1.4



+PE



+BeautifulPrompt



+Ours

An imaginative anime-style illustration that features the iconic Sydney Opera House with its distinctive white sail-like shells, sitting adjacent to the towering Eiffel Tower with its intricate iron lattice work. Both structures are set against a vibrant blue night sky, pulsating with dynamic energy, where yellow stars burst forth amidst swirling patterns of electric blue. The fantastical scene is further accentuated by the exaggerated proportions and stylized elements typical of anime art, creating a surreal and whimsical landscape.



SDv2.1



+PE



+BeautifulPrompt



+Ours

A clean white plate sits empty on a polished wooden table, with no bananas in sight. Beside it, a clear glass stands, also devoid of any orange juice, reflecting the light from the room. The table surface is smooth and the area around the plate and glass is uncluttered, emphasizing their emptiness.



SDv2.1



+PE



+MagicPrompt



+Ours

A playful monkey with a chestnut coat and bright eyes is clumsily handling a crimson red heart-shaped tea pot. The monkey sits in a verdant jungle environment, surrounded by an array of glossy green leaves and suspended vines. The tea pot, with its glossy ceramic finish, reflects the dappled sunlight that filters through the dense canopy overhead.



SD3



+PE



+BeautifulPrompt



+Ours

A peculiar tree stands in the center of a garden. Its branches are adorned with square-shaped, blue apples that hang amidst circular, bright yellow leaves. The contrast between the unconventional fruit and the vibrant foliage creates a striking visual against the backdrop of a clear sky.



DALL-E 3



+PE



+Ours

In the open expanse of a school's sports field, under the clear blue sky of a radiant sunny day, four vibrant American footballs are captured in mid-flight. The footballs, featuring hues of red, blue, yellow, and green, are spherical in shape, contrasting sharply with the green turf below. Each ball glistens in the sunlight as they arc gracefully above the field, momentarily suspended against the backdrop of a few wispy clouds.

FLUX.1schnell



+PE



+Ours



Nine differently colored labels, each featuring the iconographic representation of a Central Processing Unit, aligned neatly for visual comparison. These square icons vary in shades from vibrant red to deep blue, with the CPU symbol prominently displayed in the center. The texture of the labels appears smooth, and they are arranged in a grid pattern on a plain, light background that enhances their visibility in the illustration.

更好的生成不真实场景

FLUX.1schnell



a photo of a suitcase right of a boat

+PE



+Ours



a photo of a tie right of a baseball bat



PixArt-α



+PE



+Ours



a photo of a suitcase left of a banana



更丰富的细节表示

PixArt



+PE



+Ours



a photo of a toothbrush



a photo of a broccoli and a vase



a photo of a baseball glove



a photo of a tennis racket and a bird



2D排版智能体——Paper2Poster

Paper2Poster：将一篇学术论文（平均22.6页，约2万token含22.6张图）自动转化为单页**学术海报**（约774词，8.7张图），用于学术会议展示。。



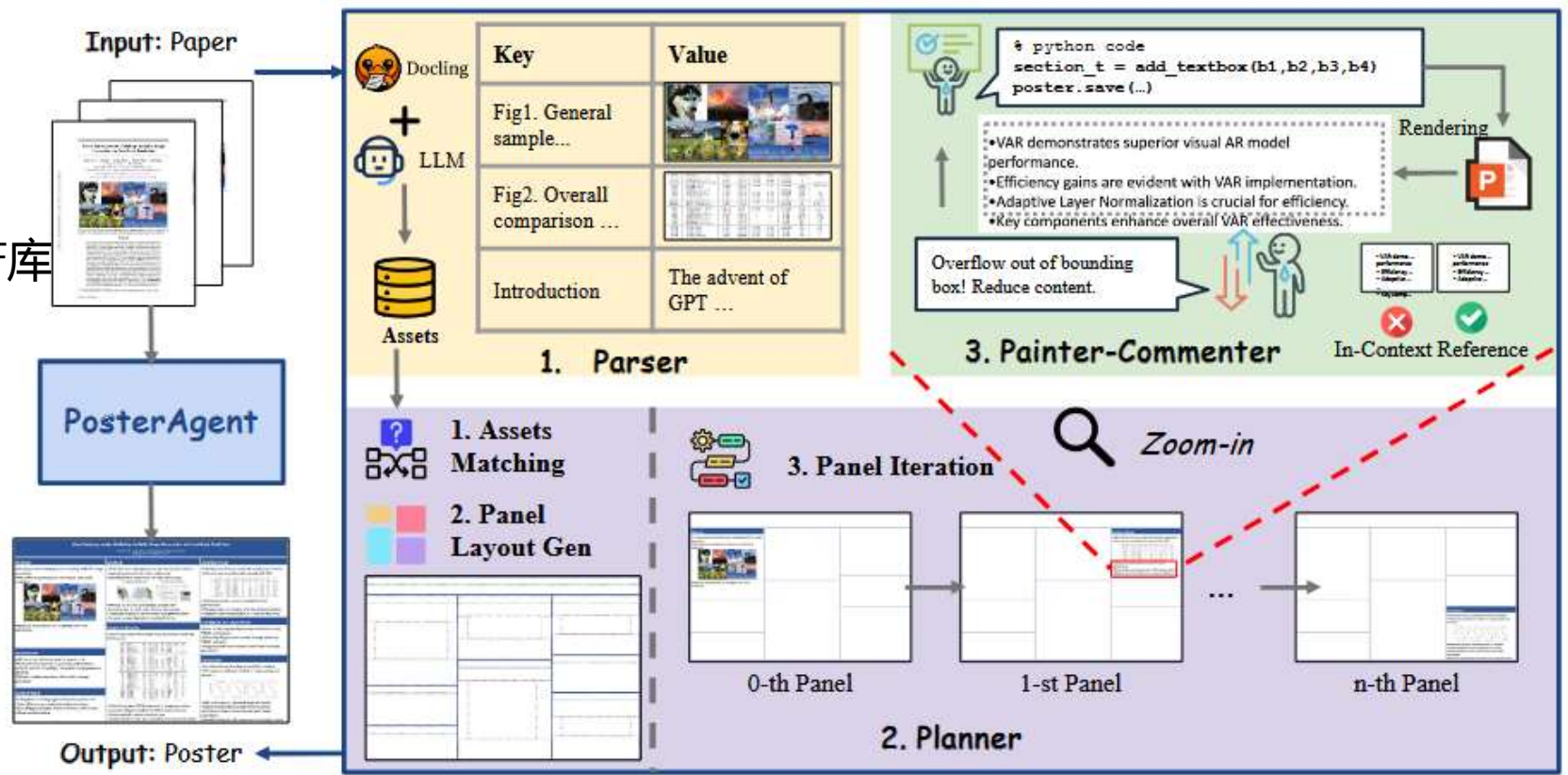
不重新训练图像模型，而是通过 Parser、Planner 和 Painter-Commenter loop 等模块完成内容提炼、二维版式规划、图文协调和可视化渲染。

整体架构

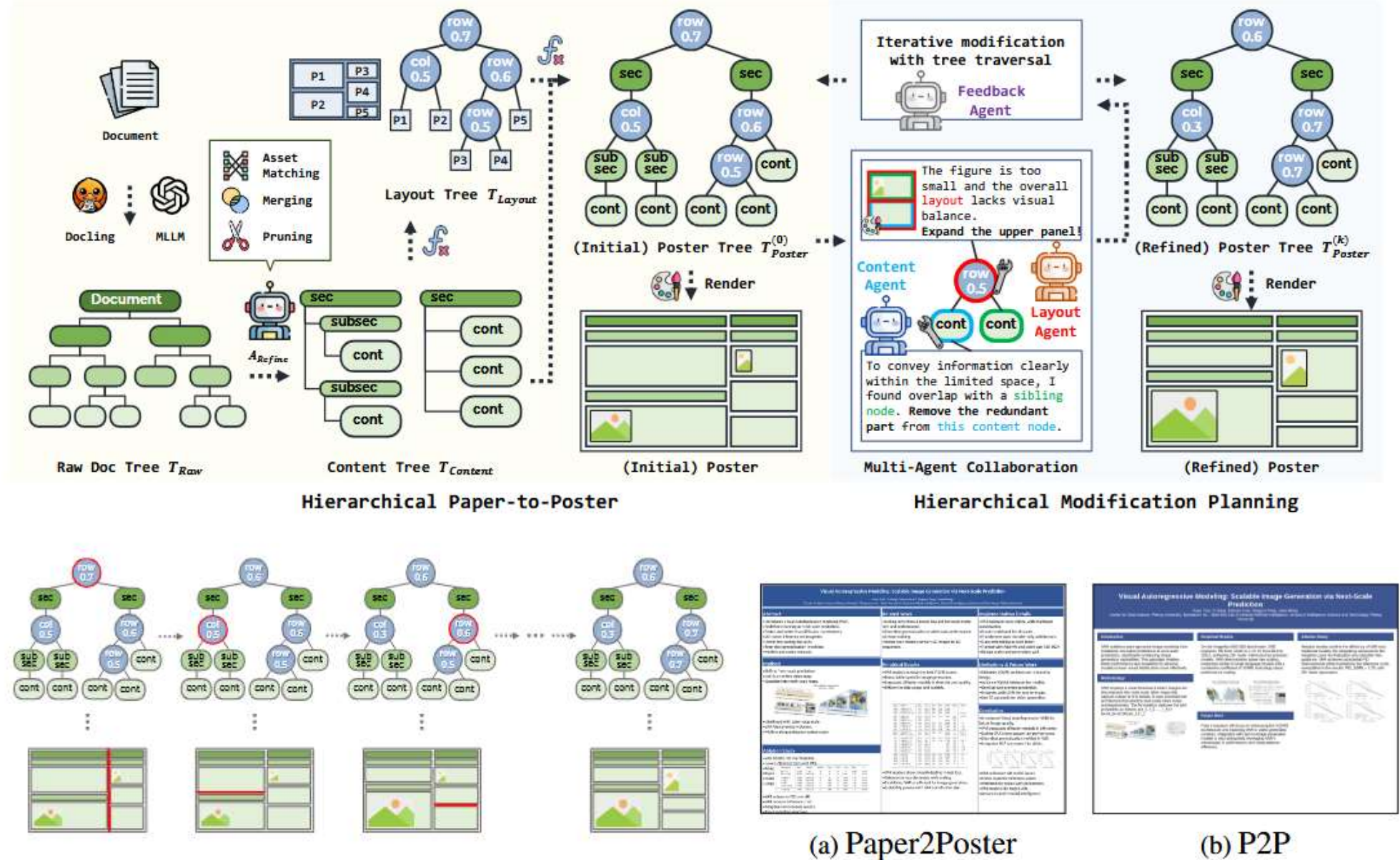
Parser：提取论文文本+图表，构建资产库

Planner：图文语义匹配+二叉树布局，保证阅读顺序与空间平衡

Painter-Commenter循环：生成面板内容 + 视觉反馈修正



迭代优化智能体——PosterForest <https://github.com/kaist-cvml/poster-forest>



PosterForest：一个**无需训练、基于层级多智能体协作**的学术海报生成框架，通过显式建模论文的层级结构（节-小节-段落）并将其与海报布局树联合优化，解决现有方法信息丢失、逻辑断裂、视觉失衡的问题。

内容 Agent 和布局 Agent 反复协调、相互反馈、逐步优化结果。它的特色在于“多 Agent 协同协作”和“迭代优化”这两个点，因为它不是一次性生成，而是通过多轮规划和修正得到更合理的 2D 结果。

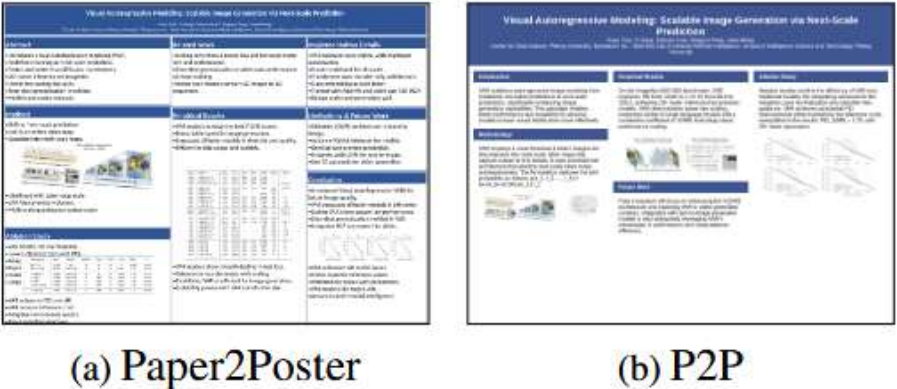


Figure 3: Poster Tree Traversal (Node-level). The

参考资料

数据集	介绍	链接
DiffusionDB	大规模 text-to-image prompt 数据集，适合做 prompt 优化、生成行为分析和图文输入研究。	https://poloclub.github.io/diffusiondb/
GenEval	面向 text-to-image alignment 的自动评测 benchmark，适合验证 prompt 优化后图像是否更符合文本意图。	https://huggingface.co/papers/2310.11513
Paper2Poster Dataset	面向 paper-to-poster 任务的配对数据与评测套件，最贴合精准 2D 排版生成与多 Agent 海报系统评估。	https://github.com/Paper2Poster/Paper2Poster

1.GenPilot: A Multi-Agent System for Test-Time Prompt Optimization in Image Generation. Findings of EMNLP 2025.
<https://aclanthology.org/2025.findings-emnlp.49/>

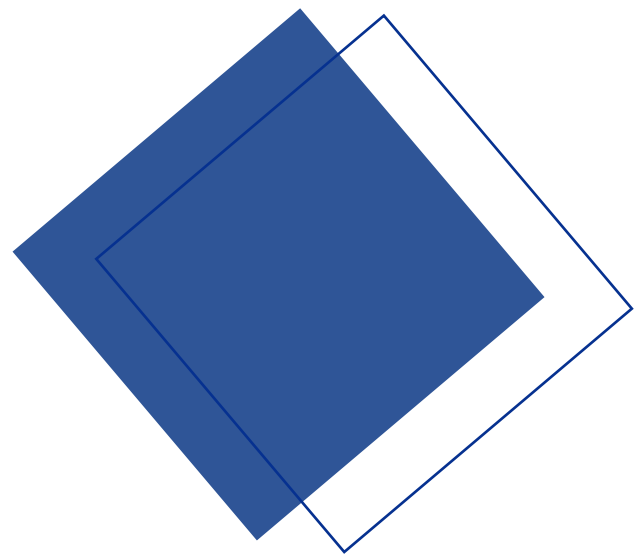
2.PromptSculptor: Multi-Agent Based Text-to-Image Prompt Optimization. EMNLP 2025 System Demonstrations.
<https://aclanthology.org/2025.emnlp-demos.59/>

3.Paper2Poster: Towards Multimodal Poster Automation from Scientific Papers. NeurIPS 2025 Datasets and Benchmarks Track.
<https://openreview.net/forum?id=p0E74lpRBD>

4.Paper2Poster: Towards Multimodal Poster Automation from Scientific Papers. arXiv 2025.
<https://arxiv.org/abs/2505.21497>

5.PosterForest: Hierarchical Multi-Agent Collaboration for Scientific Poster Generation. arXiv 2025.
<https://arxiv.org/abs/2508.21720>

6.PosterGen: Aesthetic-Aware Paper-to-Poster Generation via Multi-Agent LLMs. arXiv 2025.
<https://arxiv.org/abs/2508.17188>



02 3D生成

CAD生成——CAD-Assistant

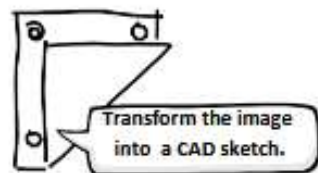
CAD-Assistant：一种工具增强型视觉语言模型（VLLM）框架。通过**工具增强范式**构建通用CAD任务求解框架，支持多模态输入和动态设计适配。

核心思路是通过 **VLLM 规划 + CAD 专用工具集 + 实时环境交互**，解决 VLLMs 在几何推理、CAD 语义理解上的短板，实现通用化 CAD 任务求解

全程**无需训练**，仅通过**复用预训练模型与工具链**，支持文本、手绘草图、3D 扫描等多模态输入，覆盖从草图参数化到逆向工程的多样化 CAD 场景。



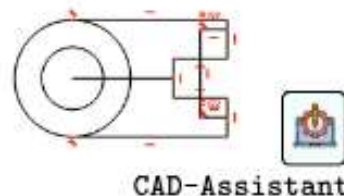
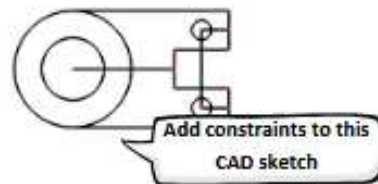
Handdrawn Image
Parameterization



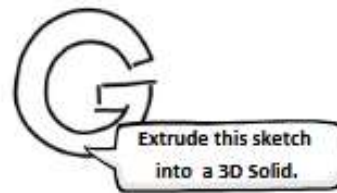
Visual Editing



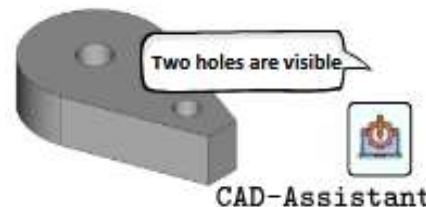
CAD Sketch
Autoconstraining



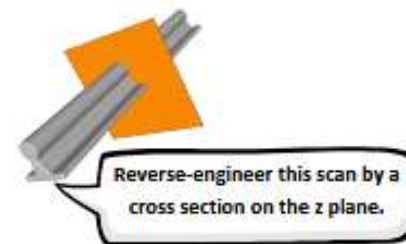
Handdrawn sketch
to 3D solid



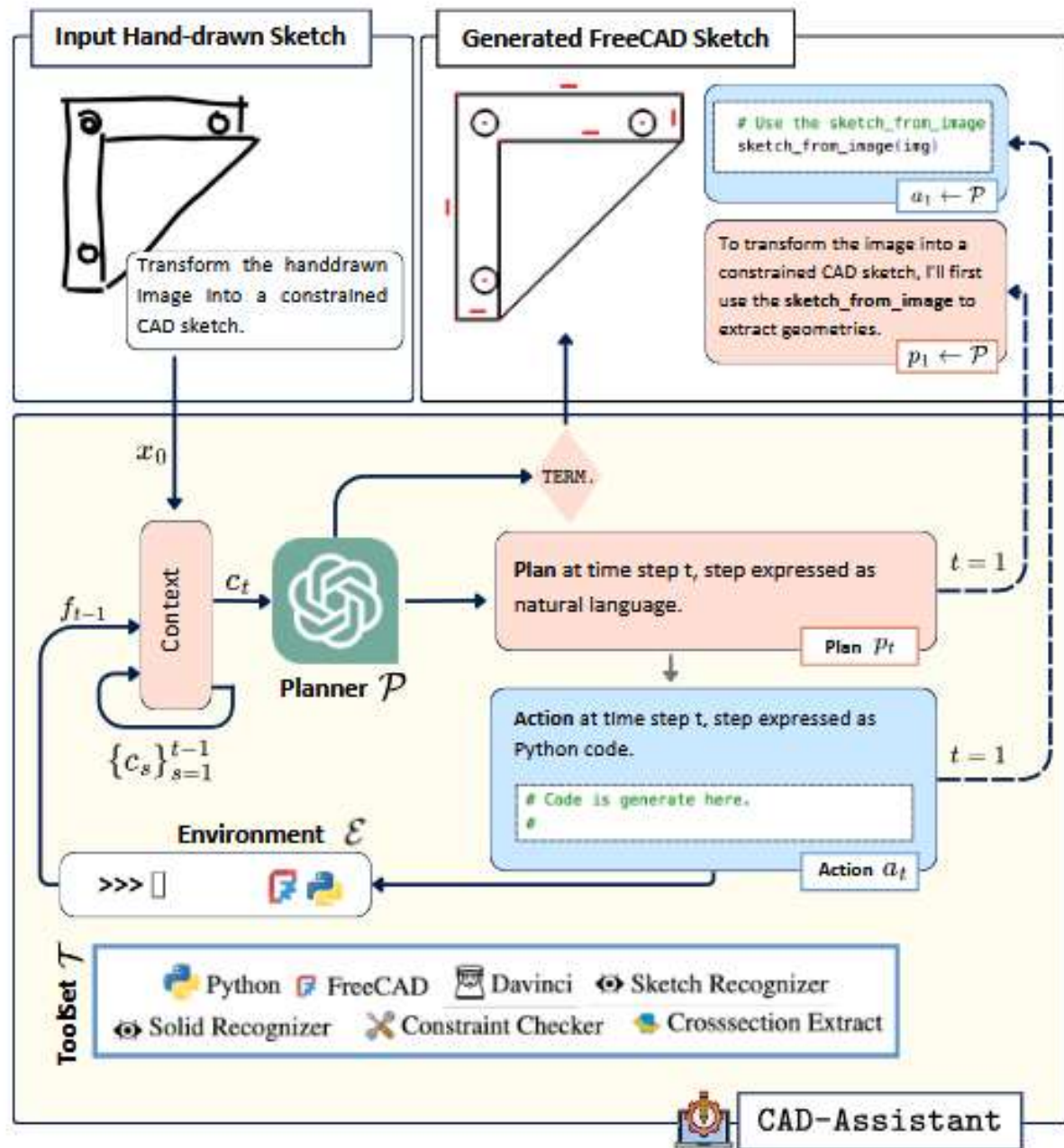
CAD Question Answering



3D Reverse Engineering



CAD生成——CAD-Assistant



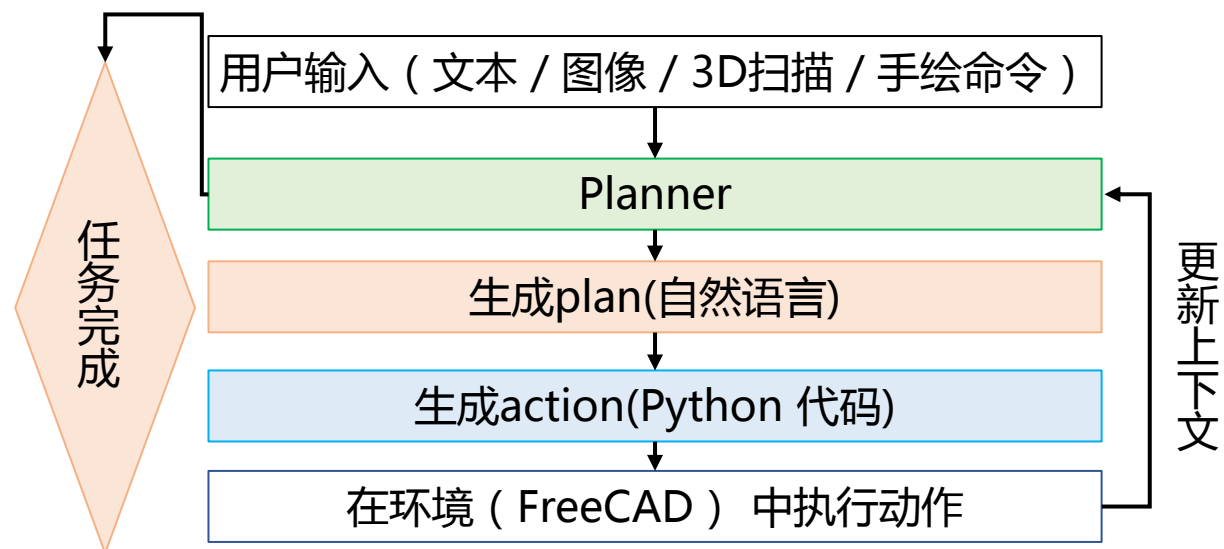
核心组件：

规划器 (Planner)：GPT-4o (做 **VLLM 大脑**)，分析用户请求 + 当前 CAD 状态，生成下一步的 Python 代码

环境 (Environment)：集成 FreeCAD 软件的 Python API，执行生成的代码

工具集 (Tool Set)：包括 FreeCAD 接口、草图参数化、渲染、约束检查、截面生成等 CAD **专用工具**

执行流程：



CAD生成——CAD-Assistant

执行示例：

Sketch to 3D Solid

USER QUERY: Create a solid by rotating this sketched profile around its axis.



Revolution Docstring

```
class Revolution:
    """
    Represents a revolution operation in FreeCAD.

    This class is part of the FreeCAD Part module and is used to create a 3D solid
    or surface by revolving a 2D sketch or profile around a specified axis.

    Attributes:
    -----
    Source (Sketch):
        The sketch object or 2D profile used as the base for the revolution.
    Axis (tuple of float):
        The axis of revolution, defined by a direction vector (default: (0.0, 1.0, 0.0)).
    Base (tuple of float):
        The base point for the axis of revolution (default: (0.0, 0.0, 0.0)).
    Angle (float):
        The angle of revolution in degrees (default: 360.0).
    Solid (bool):
        Whether the resulting object is a solid (default: True).
    AxisLink (None or Object):
        A linked object defining the axis of revolution (default: None).
    Symmetric (bool):
        Whether the revolution is symmetric about the sketch plane (default: False).
```

PLAN: I need to extract the geometric elements and constraints from the hand-drawn image.

ACTION:

```
# Use the sketch_from_image function to understand the current geometries
sketch_from_image(img)
```

ENVIRONMENT: *{{ Returns primitives in JSON format }}*

PLAN: I will create a FreeCAD sketch and add the detected geometries. Then, I'll revolve the sketch around an axis to create a 3D solid.

ACTION:

```
# Create a new FreeCAD document and sketch
doc = App.newDocument("RevolveSketch")
sketch = doc.addObject('Sketcher::SketchObject', 'Sketch')

# Add the extracted geometries to the sketch
line1 = Part.LineSegment(App.Vector(0.0078, 0.4922, 0), App.Vector(0.3359, 0.4922, 0))
line2 = Part.LineSegment(App.Vector(-0.3359, -0.1641, 0), App.Vector(0.0078, -0.1641, 0))
line3 = Part.LineSegment(App.Vector(-0.3359, -0.4922, 0), App.Vector(-0.3359, -0.1641, 0))
line4 = Part.LineSegment(App.Vector(-0.3359, -0.4922, 0), App.Vector(0.3359, -0.4922, 0))
line5 = Part.LineSegment(App.Vector(0.3359, -0.4922, 0), App.Vector(0.3359, 0.4922, 0))
line6 = Part.LineSegment(App.Vector(0.0078, -0.1641, 0), App.Vector(0.0078, 0.4922, 0))

# Add lines to the sketch
sketch.addGeometry(line1)
sketch.addGeometry(line2)
sketch.addGeometry(line3)
sketch.addGeometry(line4)
sketch.addGeometry(line5)
sketch.addGeometry(line6)

# Calculate the revolution center to achieve a 1mm radius
# The sketch is centered around the origin, so we need to offset the axis by 1mm
revolution_axis = (0.0, 1.0, 0.0) # Y-axis
revolution_base = (1.0, 0.0, 0.0) # Offset by 1mm along the X-axis

# Create the revolution
revolve = doc.addObject("Part::Revolution", "Revolve")
revolve.Source = sketch
revolve.Axis = revolution_axis
revolve.Base = revolution_base
revolve.Angle = 360.0
revolve.Solid = True

# Recompute the document to apply changes
doc.recompute()
```

ENVIRONMENT: Execution Success.



FreeCAD Model

3D场景生成——SAGE <https://nvlabs.github.io/sage/>

SAGE：基于多智能体的3D场景生成框架，能够根据任意用户文本描述，自动生成可直接部署于机器人仿真器的、物理稳定且视觉真实的室内场景，并利用这些场景规模化生成训练数据，提升具身智能策略的泛化能力。



bedroom



Fairy-tale princess room



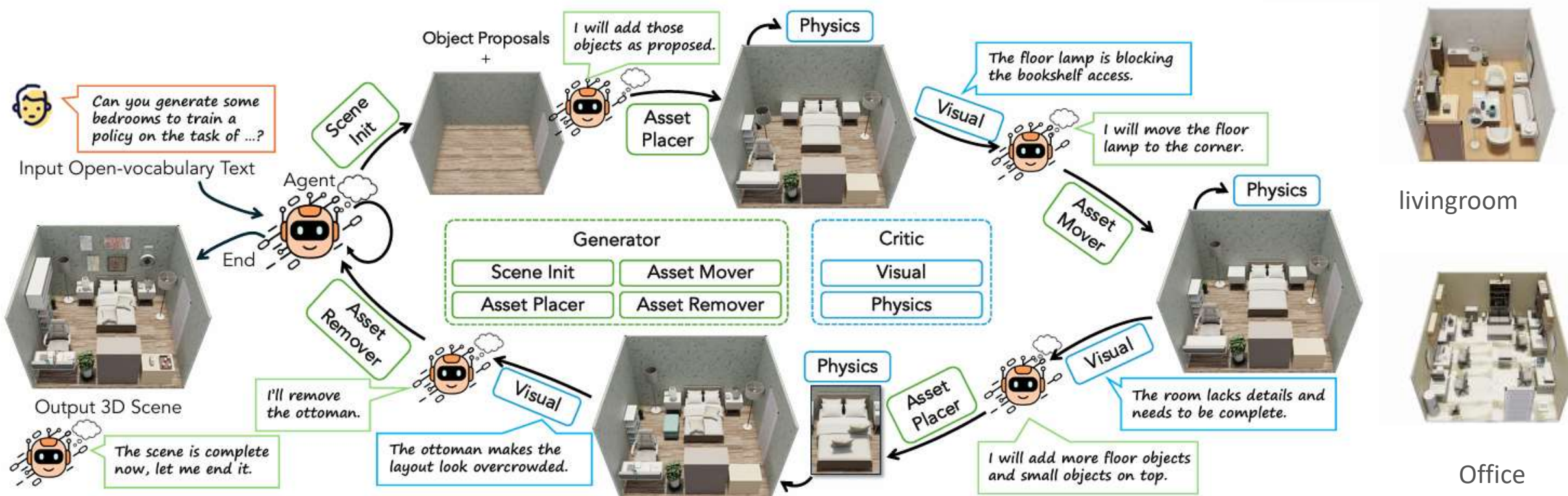
Rusty and dusty restroom



Gym

三个阶段：

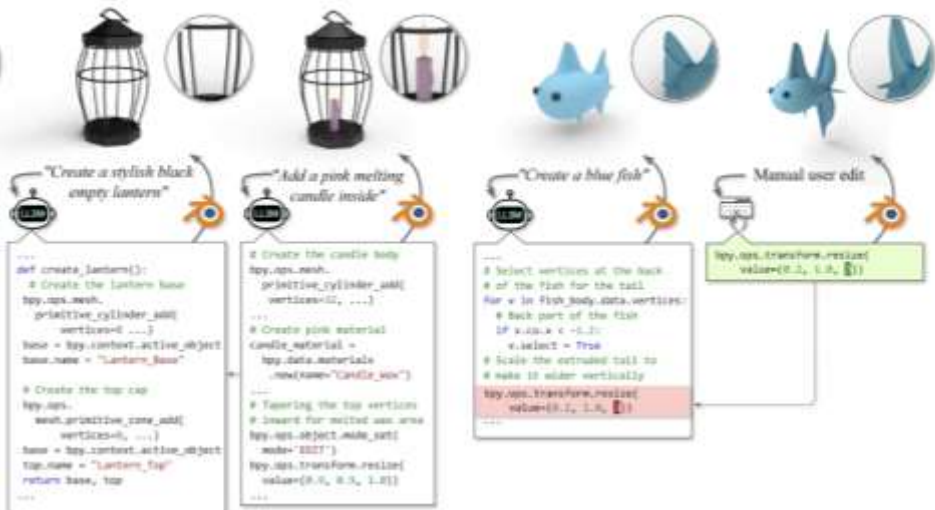
智能体驱动场景生成
场景规模化
动作生成与策略学习



3D模型生成——LL3M

LL3M：多智能体系统，利用预训练大语言模型通过编写可解释、模块化的Blender Python代码，从文本描述生成和编辑3D资产，**将3D建模重构为代码编写任务**。其中，代码是**可解释、可编辑、模块化的**，用户可通过修改代码参数或自然语言指令进行迭代式**共创**。

不是传统扩散式 text-to-3D，而是多agent 协作完成 plan / retrieve / write / debug / refine。



1.初始创建阶段

规划智能体：将用户提示分解为子任务

检索智能体：查询

BlenderRAG获取文档与示例

编码智能体：

编写并执行Blender代码

2.自动优化阶段

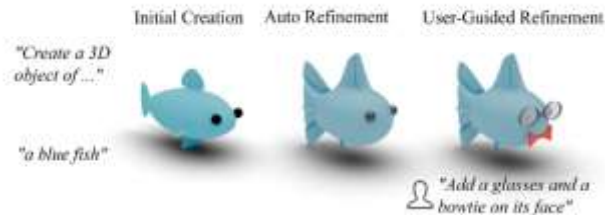
评论智能体：多视角渲染

+VLM评估视觉问题,提出修复建议

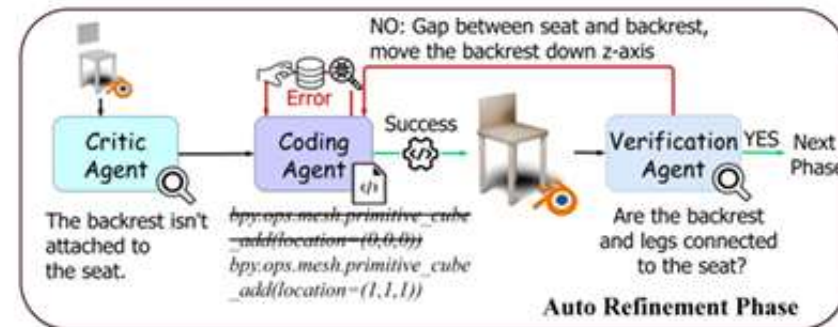
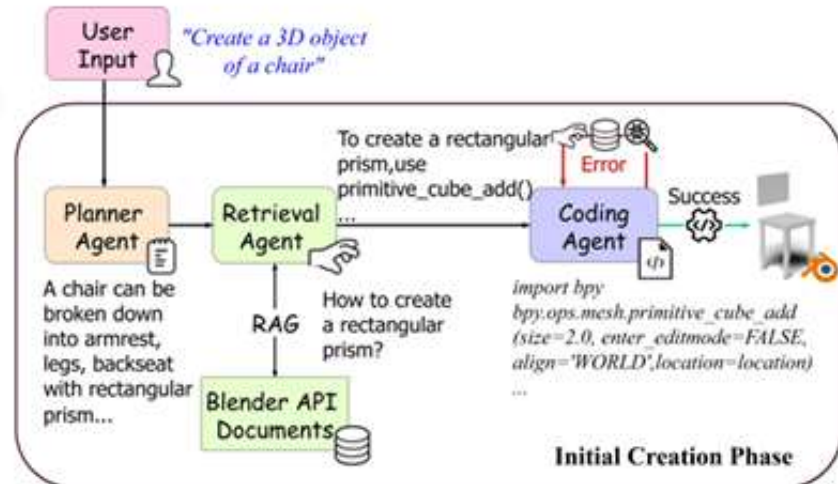
验证智能体：检查编码智能体是否正确实现修复

3.用户引导阶段

用户代理接收自然语言编辑指令，触发**编码智能体**对现有代码进行局部修改



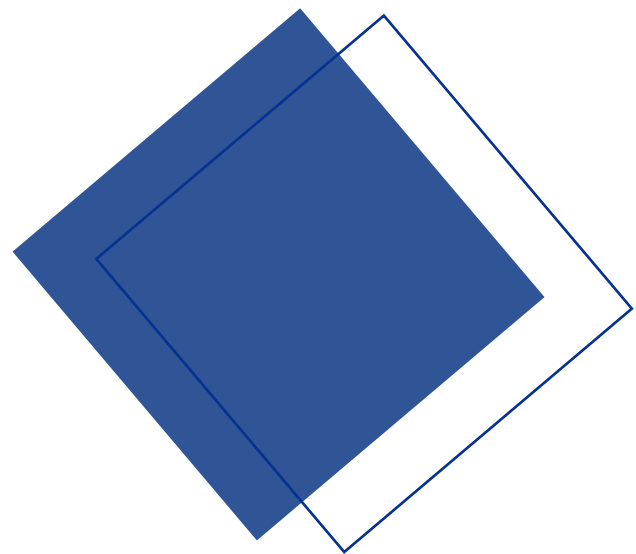
<https://threedle.github.io/ll3m/#>



参考资料

名称	网址	描述
SAGE-10k	https://huggingface.co/datasets/nvidia/SAGE-10k	SAGE 论文配套的数据集，给 embodied AI 提供 simulation-ready 的 agentic 3D scenes。包含10,000 scenes、50 种 room types/styles、565K unique 3D objects。
Eval3DAIGC-198	https://huggingface.co/yisuanwang/Idea23D	专用评测集，包含 198 个 multimodal IDEA 输入，用于评估生成出的 3D 内容与输入想法的一致性。
ScanNet / ScanNet++	https://scannetpp.mlsq.cit.tum.de/scannetpp/	ScanNet 是一个 RGB-D 视频数据集，包含超过 1500 个扫描的 250 万个视角，数据集中提供了 3D 相机姿势、表面重建和实例级语义分割标注。
DeepCAD	https://github.com/rundiwu/DeepCAD	DeepCAD 是一个 CAD 数据集，由 179,133 个模型及其 CAD 构造序列组成。可用来训练 3D 形状的生成模型。

1. Xia, Hongchi, et al. "Sage: Scalable agentic 3d scene generation for embodied ai." arXiv preprint arXiv:2602.10116 (2026).
2. Lu, Sining, et al. "LI3m: Large language 3d modelers." arXiv preprint arXiv:2508.08228 (2025).
3. Mallis, Dimitrios, et al. "CAD-assistant: tool-augmented vllms as generic cad task solvers." Proceedings of the IEEE/CVF International Conference on Computer Vision. 2025.
4. Liu, Xinhang, Chi-Keung Tang, and Yu-Wing Tai. "Worldcraft: Photo-realistic 3d world creation and customization via llm agents." arXiv preprint arXiv:2502.15601 (2025).
5. Chen, Junhao, et al. "Idea23d: Collaborative lmm agents enable 3d model generation from interleaved multimodal inputs." Proceedings of the 31st International Conference on Computational Linguistics. 2025.
6. Wen, Beichen, et al. "3d scene generation: A survey." arXiv preprint arXiv:2505.05474 (2025).
7. Liu, Jian, et al. "A comprehensive survey on 3d content generation." arXiv preprint arXiv:2402.01166 (2024).



03 视频生成

角色视频生成智能体——AniMaker

AniMaker 是一个多智能体框架，旨在高效地**从文本输入生成连贯的长篇叙事动画**。与传统生成僵硬且支离破碎的剪辑方法不同，AniMaker 支持多候选人生成、智能剪辑选择以及全局故事层面的一致性。



Once upon a time, in a small town, there was a square. In the square, there was a big sack. The sack was full of toys. It belonged to Tom, an honest boy. Tom loved to share his toys with other kids. One sunny day, Tom went to the square with his sack. He saw a little girl named Lily. Lily was sad because she had no toys to play with. Tom wanted to help her. Tom said, "Lily, do you want to play with my toys? I have a big sack of them." Lily looked at Tom and smiled. "Yes, please!" she said. Tom opened the sack and they played together all day. Tom and Lily had so much fun playing with the toys. They shared and laughed together. All the other kids in the square saw how happy they were. From that day on, Tom always brought his big sack of toys to the square. He shared them with all the kids, and they all had fun together. Tom was an honest and kind boy, and everyone loved him.



<https://animaker-dev.github.io/>

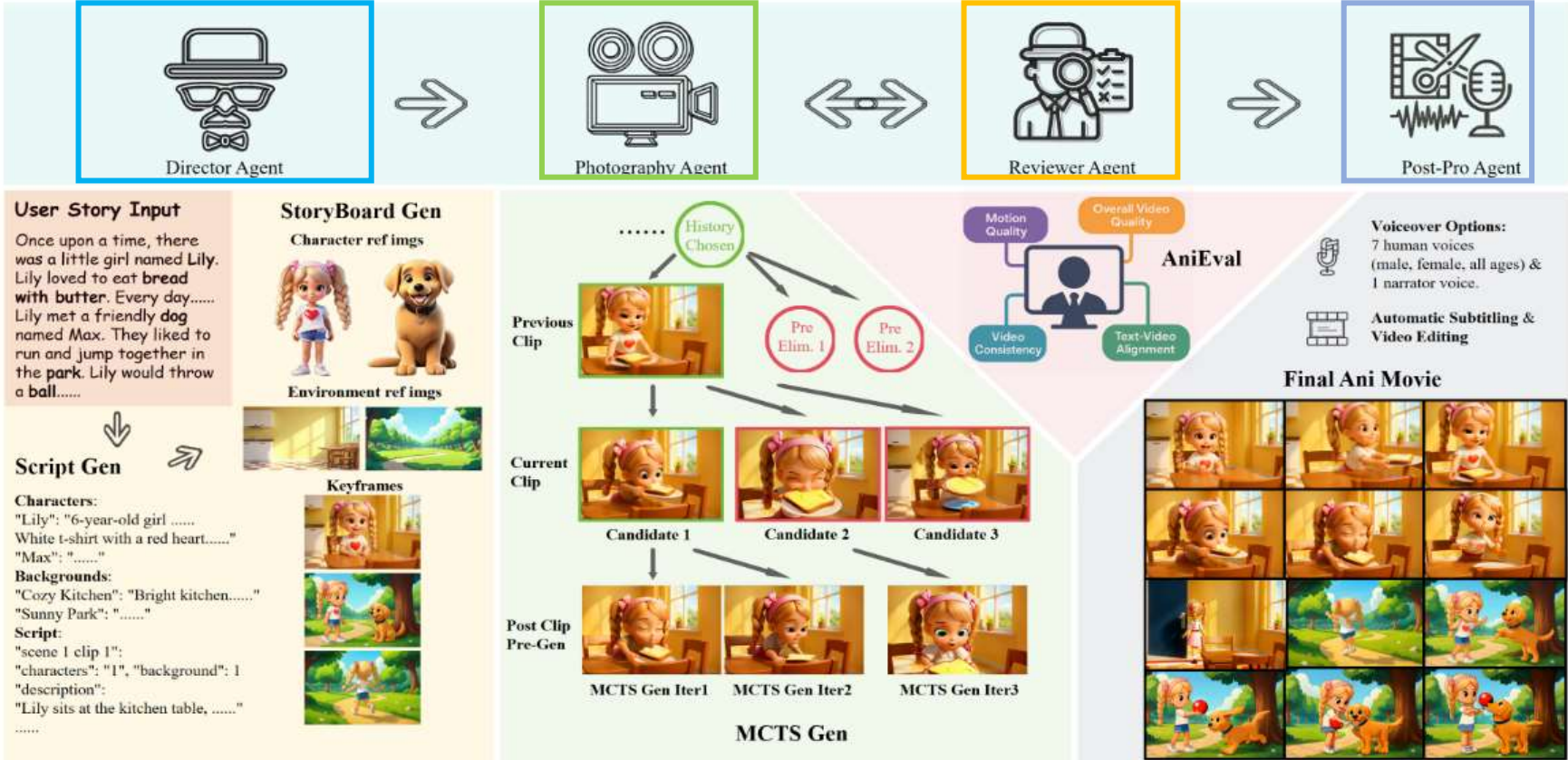
角色视频生成智能体 —— AniMaker

摄影Agent：借鉴电影制作中的 “No Good” (NG) 流程，即为达到完美镜头需录制多次镜头。提出了MCTS-Gen，受蒙特卡洛树搜索 (MCTS) 启发，高效生成多个候选片段并选择高潜力片段。

审稿Agent：提出AniEval，**多镜头动画评估框架**，用于评估故事层面的一致性、动作完成度以及片段间的动画特性。

后期制作Agent (Gemini 2.0 Flash + CosyVoice2 + MoviePy)：通过三个阶段将视频片段转化为精致的动画影片。首先生成详细的**配音脚本**，指定叙述、对话、情感基调和期望的语音音色，并评估文本长度以实现视听同步。随后生成**音频轨道**。最后组装并确保**画面、配音和字幕**之间的精确同步。

导演Agent (Gemini 2.0 Flash + Hunyuan3D + FLUX1-dev + GPT-4o)：从输入文本生成分镜，定义多场景和多角色叙事



角色视频生成智能体 ——AniMaker

Prompt参考

Below is a script for an animation. Each line describes a shot lasting approximately 5 seconds.

- {script}
- Please create a professional voice-over script following these guidelines:
- Each line corresponds to one shot from the original script, keeping appropriate length for 5-second shots (10-13 words per line). Thus your script should have exactly {script_lines_num} lines.
 - Format all lines using the pattern: {emotion:voice_type} content, where:
 - emotion is the emotional tone (e.g., happy, sad, curious)
 - voice_type is one of: narrative, boy, girl, youngman, youngwoman, woman, man, elderly
 - For narration, use voice_type "narrative"
 - For character dialogue, wrap spoken words in <speake{emotion:voice_type} >/speake> tags and use:
 - "boy" for boy characters
 - "girl" for girl characters
 - "youngman" for young male characters
 - "youngwoman" for young female characters
 - "woman" for adult female characters
 - "man" for adult male characters
 - "elderly" for elderly charactersOnly words directly spoken by characters in the script should be wrapped in <speake{emotion:voice_type} >/speake> tags. Note that {emotion:voice_type} for the spoken words must be enclosed within <speake >/speake> tags.
 - Preserve the original sequence of narration and dialogue

Example formats:

- Narration: {thoughtful:narrative} The sun cast long shadows across the empty field.
- Boy dialogue: <speake{curious:boy} >What is this tiny plant growing here?</speake>
- Mixed: {gentle:narrative} She knelt beside him and said, <speake{excited:female} >Look how much you've grown!</speake>

Use appropriate emotions that match each moment in the story.

Directly give the voice-over script without any guide words or markdown. Do not mark the serial number.

If the number of generated rows does not match, ADD:
[IMPORTANT: Your response MUST contain exactly {script_lines_num} formatted voice-over lines (one line per shot).]

Fig. 12. Prompt for Voiceover Script

Story {story}

Schema Instruction
Create a structured animation script in JSON format based on the provided story. Follow this exact schema:

```
{
  "characters": [ {
    "id": 1, "name": "[Character's full name] eg. Mia",
    "description": "[Character summary (age/gender/build)]; [Hair style & color]; [Facial features]; [Upper garment style & color]; [Lower garment style & color]; [Footwear style & color] eg. 12-year-old girl with petite frame; Chestnut hair in messy braid; Round face with freckled cheeks; Oversized mustard yellow sweater; Patched navy corduroy overalls; Scuffed brown ankle boots." },
    "environments": [ {
      "id": 1, "name": "[Unique environment name] eg. Moonlit Alley",
      "description": "[A vivid and cinematic depiction of the scene, encompassing the time of day, weather conditions, primary elements, colors, and ambient details. The description should be sufficiently detailed to translate effortlessly into a visual masterpiece, with distinct visual components for the background, midground, and foreground.] eg. Moonlit cobblestone alley with glistening rain puddles, Victorian gas lamps casting amber pools of light, distant church bell tolling midnight." },
      "script": [ {
        "scene": 1, "environment": 1,
        "clips": [ {
          "id": 1, "characters": [ "1" ],
          "description": "[A concise and detailed description of the movie clip, including character actions, facial expressions, dialogue, interactions with the environment, and camera shots.] eg. Jimmy sits alone in a dim room, eyes fixed on a worn photograph. A soft sigh escapes his lips as he gently traces the faces with a finger. \\\"I miss you,\\\" he whispers, his voice tinged with sorrow. The camera slowly zooms in, focusing on his face, capturing the sadness in his eyes." } ] ] ] ] }
```

The following rules must be adhered to:

- Include 2, 3, or 4 #environments# and 1, 2, or 3 #characters# with detailed #descriptions#.
- Animals can be #characters# and may talk, but their #character descriptions# must be natural, not anthropomorphic. Plants cannot be #characters#.
- An #environment# is a background for the story. The #environment description# must not include characters or plot elements and should be easy to visualize.
- A new #scene# starts only when the #environment# changes. An #environment# can be reused in different #scenes#.
- Each #scene# must contain sequential clips that include:
 - #Environment# reference
 - Present #characters#
 - Visual actions (camera moves, transitions)
 - Natural dialogue
- A maximum of 2 #characters# are allowed in a single #clip#.
- Include #clips# showing characters entering or exiting an #environment#. For example: "Jimmy stood up with a look of disappointment, turned back and walked away, the camera zoomed out."
- Use the present tense for all action descriptions.
- Return the content in JSON format only, without any introductory text or markdown formatting.

Fig. 7. Prompt for Story to Script Generation

Generate a prompt for image generation that describes this clip:
{clip[description]}

The prompt must mention these characters: {', '.join(character_names)} and this environment: {env_name}. Note that the prompt should depict a static image rather than a dynamic action sequence. And it should be as simple as possible. Do not include dialogue.

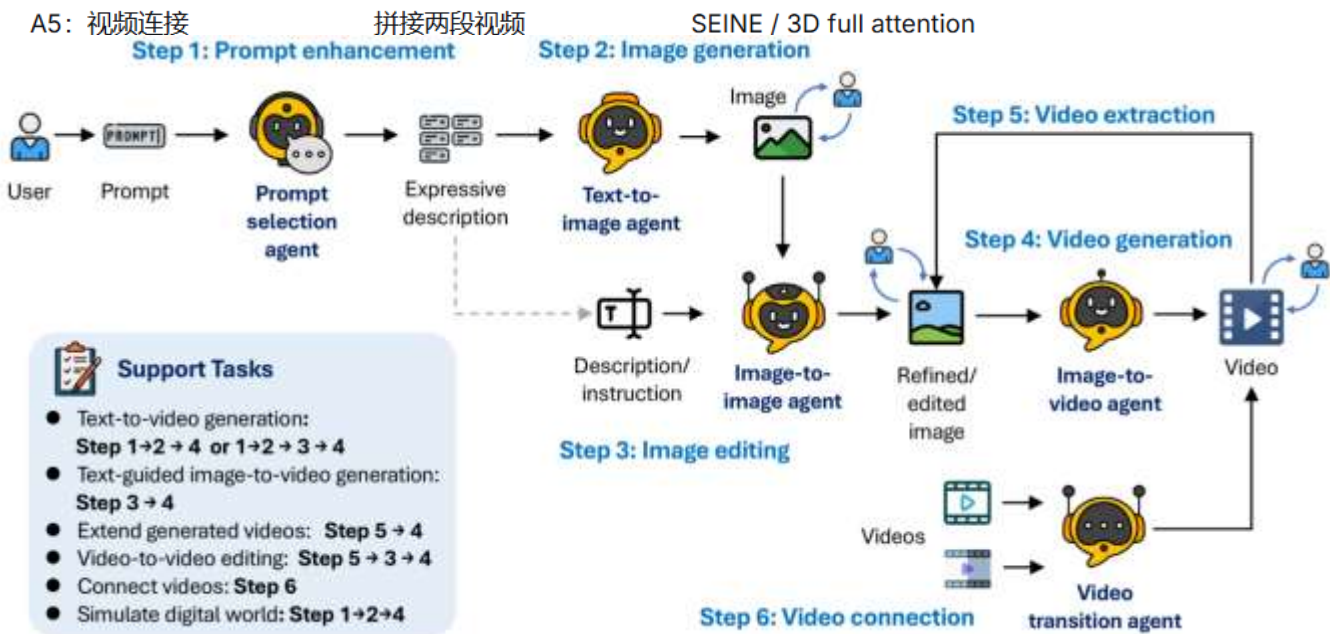
Format the prompt like:
"16-9, image generation: [{Character 1} (in image 1) (and [{Character 2} in (in image 2)) is/are doing something in the {Environment} (in image x)]"
e.g
1."16-9, image generation: The little boy (in image 1) walks towards a rose under the frost-kissed glass dome in the icy desert night (in image 2), looking worriedly."
2."16-9, image generation: The little boy (in image 1) met the fox (in image 2) in the rose garden (in image 3), and they exchanged warm greetings."

Fig. 9. Prompt for Scene Image Generation

分镜视频生成智能体——Mora

Mora：基于多智能体协作的开源视频生成框架，目标是用模块化方案复现并对标闭源模型 Sora 的能力，通过智能体分工、自调制微调、无数据训练与人在环路筛选，实现高质量、多任务的视频生成。

智能体	功能	使用模型
A1: 提示选择与生成	增强用户文本提示	GPT-4 / LLaMA
A2: 文本生成图像	生成首帧	Stable Diffusion XL / SD3
A3: 图像生成图像	编辑图像	InstructPix2Pix / SD3
A4: 图像生成视频	生成动态视频	Stable Video Diffusion / Open-Sora-Plan



Video-to-video editing: Change the setting to the 1920s with an old school car. make sure to keep the red color.

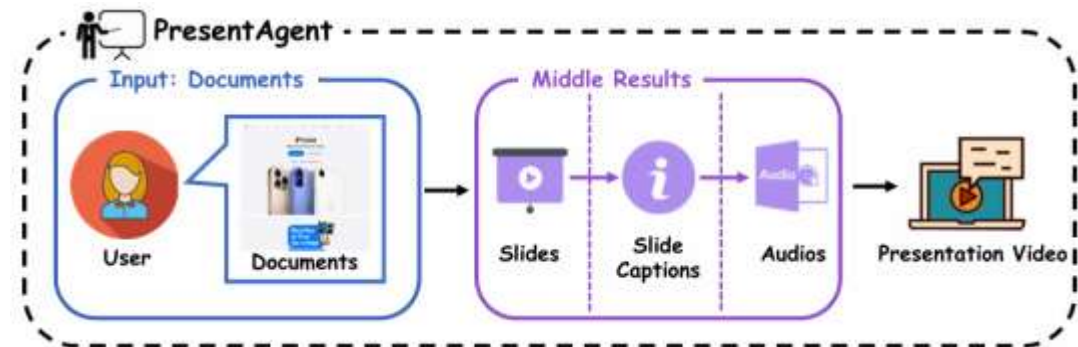
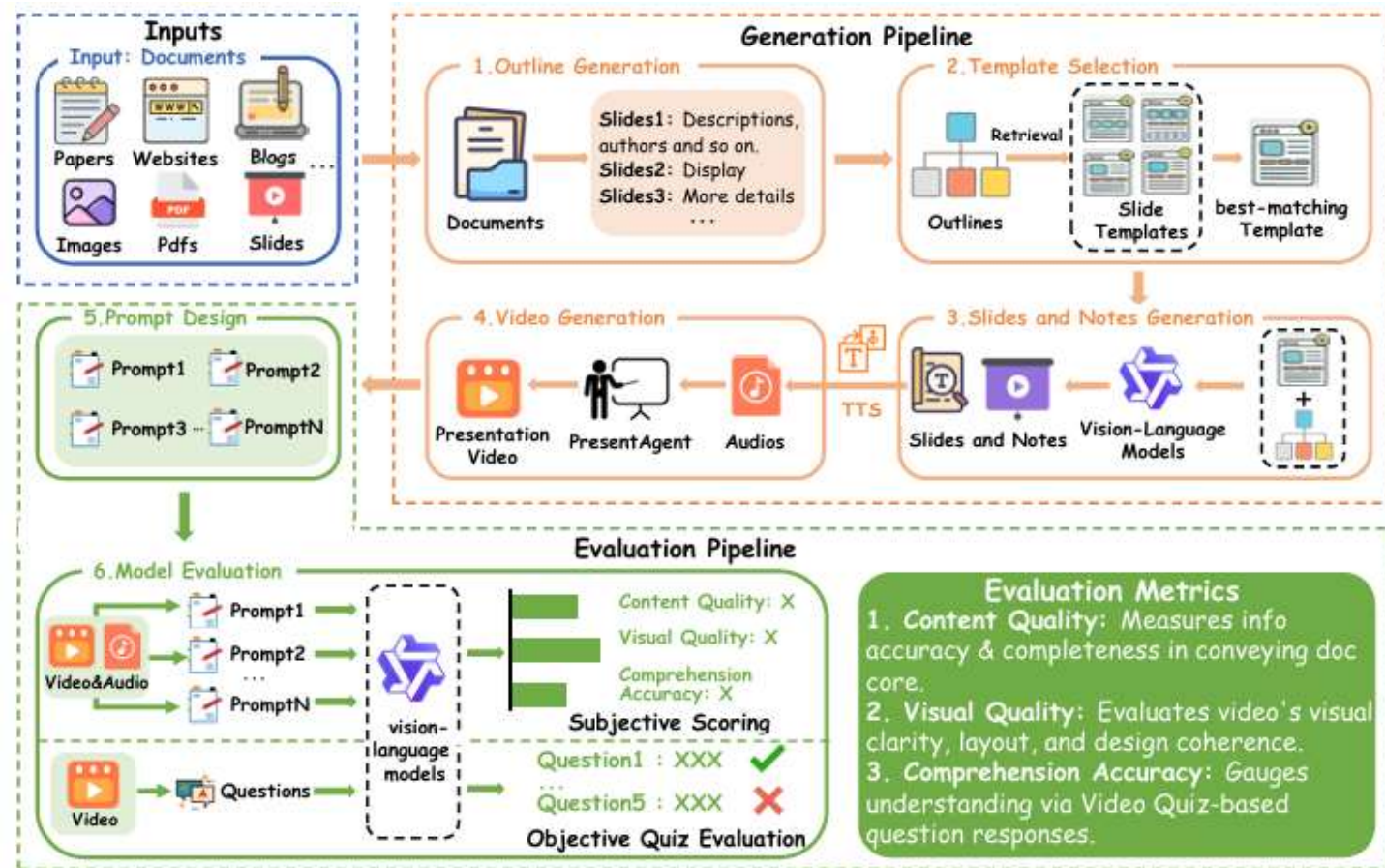
Text-to-video generation: A majestic mountain range covered in snow, with the peaks touching the clouds and a crystal-clear lake at its base, reflecting the mountains and the sky, creating a breathtaking natural mirror.



演示视频生成智能体——PresentAgent

PresentAgent：多模态智能体框架，将长篇文档（如论文、网页、报告）自动转化为带解说、幻灯片结构的演示视频，接近人类制作水平。完成了**视觉+语音+时间轴**三者的同步对齐。

- 文档解析
- 幻灯片生成
- 旁白生成
- 音视频合成



Demo : Slides and captions



Hello everyone and welcome to the presentation. My name is [Your Name], and I'm delighted to be here with you today to discuss the topic you see on the screen. Let's dive right into our discussion.



This slide illustrates the parallelization workflow, where multiple LLMs process an input simultaneously. We then aggregate their outputs, either by sectioning the task for speed or using a voting system to improve confidence in the final result.



The fundamental building block of an agent system is a Large Language Model augmented with external capabilities. This allows the model to actively query for information, utilize tools, and manage memory, moving beyond simple text generation to dynamic problem-solving.



The evaluator-optimizer workflow uses two LLMs in a feedback loop. One generates a response, while the other critiques it for iterative refinement. This process is highly effective for tasks with clear criteria, such as literary translation or complex search.

Videos



The fundamental building block of an agent system is a Large Language Model augmented with external capabilities for queries, tool use, and data operations.



Processes an input with multiple LLMs simultaneously. The separate outputs are then combined programmatically to produce a single, unified result.

参考资料

名称	网址	描述
Doc2Present	https://huggingface.co/datasets/AIGeeksGroup/Doc2Present	多领域、多文体的真实对照数据集，其中每对数据都包含一个文档与一个配套的演示视频。数据包括：商业报告、产品手册、政策简报、教程类文档等。
VidGen-1M	https://huggingface.co/datasets/Fudan-FUXI/VIDGEN-1M	文本转视频模型训练数据集。该数据集通过从粗到细的策划策略生成，保证高质量视频和详细的字幕 时间一致性极佳。
Panda-70M	https://github.com/snap-research/Panda-70M	7000 万对高质量视频字幕对。
MiraData	https://github.com/mira-space/Miradata	具有长时长和结构化字幕的大规模视频数据集
MMTrail	https://github.com/litwellchi/MMTrail	包含超过2000万段预告片，具备高质量多模态字幕，整合上下文、视觉帧和背景音乐

1. Yuan, Zhengqing, et al. "Mora: Enabling generalist video generation via a multi-agent framework." arXiv preprint arXiv:2403.13248 (2024).
2. Shi, Jingwei, et al. "Presentagent: Multimodal agent for presentation video generation." Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: System Demonstrations. 2025.
3. Shi, Haoyuan, et al. "AniMaker: Multi-Agent Animated Storytelling with MCTS-Driven Clip Generation." Proceedings of the SIGGRAPH Asia 2025 Conference Papers. 2025.
4. Huang, Kaiyi, et al. "Genmac: compositional text-to-video generation with multi-agent collaboration." Proceedings of the AAAI Conference on Artificial Intelligence. Vol. 40. No. 7. 2026.
5. He, Liu, et al. "Kubrick: Multimodal agent collaborations for synthetic video generation." arXiv preprint arXiv:2408.10453 (2024).
6. Mu, Lingzhou, et al. "FantasyHSI: Video-Generation-Centric 4D Human Synthesis in Any Scene Through a Graph-Based Multi-Agent Framework." Proceedings of the AAAI Conference on Artificial Intelligence. Vol. 40. No. 10. 2026.
7. Wu, Kuizong, et al. "AutoMV: An Autonomous Agent Framework for Real Estate Marketing Video Generation." Proceedings of the AAAI Conference on Artificial Intelligence. Vol. 39. No. 28. 2025.
8. Soni, Achint, et al. "Videoagent: Self-improving video generation for embodied planning." NeurIPS 2025 Workshop on Bridging Language, Agent, and World Models for Reasoning and Planning. 2025.

给分标准

- 内容：
扩展部分为开放式，由同学**自选Agent+图形学相关课题**完成，包括但不限于2D生成、视频生成、3D生成等。
- 基础要求：
 - 任意**Agent+图形学相关课题**。
 - 提交报告PDF，含代码链接。
 - 提交汇报PPT，按时进行汇报。
- 进阶要求：
 - 发挥想象力，展示创新，包括但不限于新应用、新方法。
 - 具体实现不做限制，实现方式不影响给分，推荐使用免费资源（详见下页）。
 - 推荐参考他人论文、博客与开源代码，但需要给出明确的引用。
- 给分标准：
 - 基础给分：10分，准时提交报告PDF和汇报PPT，按时汇报即可。
 - 质量给分：10分，根据报告创新性和汇报展示综合给分。
 - **如果包含原创性的、有效的创新，可获得整个PJ3满分（20分）。**

语言模型参考

1. 国外免费API：

- NVIDIA（免费API）：<https://build.nvidia.com/explore/discover>
- Google AI Studio（免费API）：<https://aistudio.google.com>
- OpenRouter（免费API）：<https://openrouter.ai/docs/api-reference/limits>
- Opencode（免费API）：<https://opencode.ai/zen>

2. 国内免费API：

- GLM（免费API）：<https://open.bigmodel.cn/>
- 硅基流动（免费API）：<https://siliconflow.cn/>
- Minimax（免费额度）：<https://platform.minimaxi.com/>
- KIMI（免费额度）：<https://platform.kimi.com/>

3. 本地部署：

- Ollama: <https://ollama.com/>
- LMStudio: <https://lmstudio.ai/>

* 以上内容仅供参考，具体实现不做限制，实现方式不影响给分。

Agent框架参考

1. 开源Agent框架：

- DeepAgent: <https://github.com/langchain-ai/deepagents>
- OpenClaw: <https://github.com/openclaw/openclaw>
- Hermes Agent: <https://github.com/NousResearch/hermes-agent>
- Claude Code: <https://github.com/anthropics/claude-code>
- Codex: <https://github.com/openai/codex>
- Gemini Cli: <https://github.com/google-gemini/gemini-cli>

2. 开源图形学软件：

- ComfyUI: <https://www.comfy.org/zh-cn/>
- Blender: <https://docs.blender.org/api/current/index.html>
- FreeCAD: https://wiki.freecad.org/Python_scripting_tutorial/zh-cn

* 以上内容仅供参考，具体实现不做限制，实现方式不影响给分。

报告要求

- 要求：使用英文，描述清晰，提交PDF文件。
- 格式：长度格式不做限制，推荐使用latex，参考ICLR会议模板。
(<https://github.com/ICLR/Master-Template/>)
- 内容：
 - ✓ 摘要
 - ✓ 介绍
 - ✓ 相关工作
 - ✓ 方法
 - ✓ 实验结果
 - ✓ 结论
- 注意：
 - ✓ 如果使用公开代码和数据需要引用。
 - ✓ 请在摘要附上代码链接。
 - ✓ 请在结尾给出具体组员的姓名、学号与分工情况。

汇报要求

- 制作汇报PPT，讲解如下内容：
 - ✓ 选题的内容
 - ✓ 方法创新性
 - ✓ 结果展示
 - ✓ 人员分工
- 时间：6月11日（15周） 6月18日（16周）
- 内容：展示内容可以是图片、视频、网站、软件等，不做限制。
- 时长：每组时间控制在10分钟左右。
- 压缩包标题：2026图形学Project3 姓名1 姓名2 姓名3 姓名4（最多4人）
- 提交到elearning：报告PDF+汇报PPT打包成zip，DDL：6月10日晚23：59